

$$\textcircled{1} \quad \Delta_2 = \begin{vmatrix} 2 & -3 \\ 4 & 5 \end{vmatrix} = 2 \cdot 5 - (-3) \cdot 4 = 10 + 12 = 22$$

$\textcircled{2}$

$$\Delta_2 = \begin{vmatrix} -5 & -6 \\ 1 & 8 \end{vmatrix} = (-5) \cdot 8 - (-6) \cdot 1 = -40 + 6 = -34$$

$\textcircled{3}$

$$\begin{aligned} \Delta_3 &= \begin{vmatrix} 1 & 2 & 3 \\ 1 & 4 & 7 \\ 2 & 6 & 13 \end{vmatrix} = 1 \cdot 4 \cdot 13 + 1 \cdot 6 \cdot 3 + 2 \cdot 7 \cdot 2 - \\ &\quad - 3 \cdot 4 \cdot 2 - 1 \cdot 6 \cdot 7 - 1 \cdot 2 \cdot 13 = \\ &= 52 + 18 + 28 - 24 - 42 - 26 = 98 - 92 = 6 \end{aligned}$$

④

$$\Delta_3 = \begin{vmatrix} 11 & 5 & 6 \\ 1 & -2 & -3 \\ 7 & 4 & 4 \end{vmatrix} = 11 \cdot (-2) \cdot 4 + 5 \cdot (-3) \cdot 7 + 1 \cdot 4 \cdot 6 - \\ - 6 \cdot (-2) \cdot 7 - 11 \cdot 4 \cdot (-3) - 5 \cdot 1 \cdot 4 = \\ = -88 - 105 + 24 + 84 + 132 - 20 = \\ = -213 + 240 = 27$$

⑤ $\Delta_3 = \begin{vmatrix} 3 & 9 & 1 \\ 7 & 12 & 5 \\ 2 & -3 & -2 \end{vmatrix} = a_{11} \cdot A_{11} + a_{21} \cdot A_{21} + a_{31} \cdot A_{31} =$

no condusy 1

$$\begin{aligned}
 &= -3 \cdot (-1)^{1+1} \cdot M_{11} + 7 \cdot (-1)^{2+1} \cdot M_{21} + 2 \cdot (-1)^{3+1} \cdot M_{31} = \\
 &= 3 \cdot \begin{vmatrix} 12 & 5 \\ -3 & -2 \end{vmatrix} - 7 \cdot \begin{vmatrix} 9 & 1 \\ -3 & -2 \end{vmatrix} + 2 \cdot \begin{vmatrix} 9 & 1 \\ 12 & 5 \end{vmatrix} = \\
 &= 3(-24 - (-15)) - 7(-18 - (-3)) + 2(45 - 12) = \\
 &= 3 \cdot (-9) - 7 \cdot (-15) + 2 \cdot 33 = -27 + 105 + 66 = 144
 \end{aligned}$$

$$\Delta_3 = \begin{vmatrix} 3 & 9 & 1 \\ 7 & 12 & 5 \\ 2 & -3 & -2 \end{vmatrix} = \begin{matrix} 3 \cdot 12 \cdot (-2) + 7(-3) \cdot 1 + 2 \cdot 9 \cdot 5 - \\ -2 \cdot 12 \cdot 1 - 7 \cdot 9 \cdot (-2) - 3 \cdot 5 \cdot (-3) = \\ = -72 - 21 + 90 - 24 + 126 + 45 = \\ = 144 \end{matrix}$$

6

$\Delta_3 =$

no spoke 1

a_{11}	a_{12}	a_{13}
1	2	3
9	5	7
8	4	3

Hand-drawn diagram of a 3x3 grid. The columns are labeled a_{11} , a_{12} , and a_1 at the top. The grid contains the following numbers:

1	2	3
9	5	7
8	4	3

The grid is circled, and there are additional vertical and horizontal lines drawn through it.

по spoke 1

$$1+1 = 2 - \text{Zerho}$$

$$1+2=3 - \text{Leereset}$$

$1+3$

merges $\downarrow$ cap. $\downarrow$ cons.

050252

через 45

$$= -13 - 58 + 12 = -59$$

7

$$\Delta_4 = \begin{vmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{vmatrix} = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} + a_{14}A_{14} =$$

no cross 1

$$= 0 \cdot (-1)^{1+1} M_{11} + 1 \cdot (-1)^{1+2} M_{12} + 1 \cdot (-1)^{1+3} M_{13} + 1 \cdot (-1)^{1+4} M_{14}$$

$$= - \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} - \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = -1 + (-1) - 1 = -3$$

$$\Delta_3 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 1 \cdot 0 \cdot 0 + 1 \cdot 1 \cdot 1 + 1 \cdot 1 \cdot 1 - 1 \cdot 0 \cdot 1 - 1 \cdot 1 \cdot 1 - 1 \cdot 1 \cdot 0 = 2 - 1 = 1$$

$$\Delta_3 = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = \underset{\circ}{1}\cancel{1}0 + 111 + \underset{\circ}{1}\cancel{0}1 - 111 - \underset{\circ}{1}\cancel{0}0 - 111 =$$

$$= 1 - 2 = -1$$

$$\Delta_3 = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = 111 + \underset{\circ}{1}\cancel{0}0 + 111 - 111 - \underset{\circ}{1}\cancel{0}1 - \underset{\circ}{1}\cancel{0}1 =$$

$$= 2 - 1 = 1$$

8

$$\Delta_4 = \begin{vmatrix} -2 & -5 & -1 & 3 \\ 2 & -5 & 9 & 1 \\ 3 & -1 & 5 & -5 \\ 2 & 18 & -7 & -10 \end{vmatrix} = a_{12} A_{12} + a_{22} A_{22} + a_{32} A_{32} + a_{42} A_{42}$$

no crossing
2

$$= (-5)(-1)^{1+2} M_{12} + (-5)(-1)^{2+2} M_{22} + (-1)(-1)^{3+2} M_{32} + 18(-1)^{4+2} M_{42} =$$

$$= 5 \begin{vmatrix} 2 & 9 & 1 \\ 3 & 5 & -5 \\ 2 & -7 & -10 \end{vmatrix} - 5 \begin{vmatrix} -2 & -1 & 3 \\ 3 & 5 & -5 \\ 2 & -7 & -10 \end{vmatrix} + \begin{vmatrix} -2 & -1 & 3 \\ 2 & 9 & 1 \\ 2 & -7 & -10 \end{vmatrix} + 18 \begin{vmatrix} -2 & -1 & 3 \\ 2 & 9 & 1 \\ 3 & 5 & -5 \end{vmatrix}$$

①

$$\Delta_3 = \begin{vmatrix} 2 & 9 & 1 \\ 3 & 5 & -5 \\ 2 & -7 & -10 \end{vmatrix} = 2 \cdot 5 \cdot (-10) + 1 \cdot 3 \cdot (-7) + 2 \cdot 9 \cdot (-5) -$$

$$- 1 \cdot 5 \cdot 2 - 2 \cdot (-5) \cdot (-7) - 3 \cdot 9 \cdot (-10) =$$

$$= -100 - 21 - 90 - 10 - 70 + 270 =$$

$$= -291 + 270 = -21$$

$$\Delta_3 = \begin{vmatrix} -2 & -1 & 3 \\ 3 & 5 & -5 \\ 2 & -7 & -10 \end{vmatrix} = (-2) \cdot 5 \cdot (-10) + 2 \cdot (-1) \cdot (-5) + 3 \cdot 3 \cdot (-7) -$$

$$- 2 \cdot 5 \cdot 3 - (-2) \cdot (-5) \cdot (-7) - 3 \cdot (-1) \cdot (-10) =$$

$$= 100 + 10 - 63 - 30 + 70 - 30 = 180 - 123 = 57$$

$$\Delta_3 = \begin{vmatrix} -2 & -1 & 3 \\ 2 & 9 & 1 \\ 2 & -7 & -10 \end{vmatrix} = (-2)9(-10) + 2 \cdot 3(-7) + 2 \cdot 1(-1) -$$

$$- 2 \cdot 9 \cdot 3 - 1(-2)(-7) - 2(-1)(-10) =$$

$$= 180 - 42 - 2 - 54 - 14 - 20 = 180 - 132 = 48$$

$$\Delta_3 = \begin{vmatrix} -2 & -1 & 3 \\ 2 & 9 & 1 \\ 3 & 5 & -5 \end{vmatrix} = (-2)9(-5) + 2 \cdot 5 \cdot 3 + 3(-1) \cdot 1 -$$

$$- 3 \cdot 9 \cdot 3 - (-2)5 \cdot 1 - 2(-1)(-5) =$$

$$= 90 + 30 - 3 - 81 + 10 - 10 = 36$$

① $\rightarrow$

$$\textcircled{=} 5 \cdot (-21) - 5 \cdot 57 + 48 + 18 \cdot 36 = -105 - 285 + 48 + 360$$

$$+ 240 + 48 =$$

$$= 696 - 390 = 306$$